

SOFTWARE DEVELOPER ROADMAP

From Zero to Big Tech Software Engineer

Phase 1 – Programming Fundamentals (0–2 Months)

Learn Java

- Variables & Data Types
- Operators
- Loops & Conditions
- Methods
- Arrays & Strings
- Object-Oriented Programming (OOP)
- Exception Handling
- Collections Framework
- File Handling

Resources

 **Java – Telusko**https://www.youtube.com/watch?v=bm0QyhwFDuY&list=PLsyebzWxl7pe_liTfNyr55kwJPWbgxB5

 **OOP – Kunal Kushwaha**https://www.youtube.com/watch?v=BSVKUk58K6U&list=PL9gnSGHSqcn01G3XjUbWzXHL8_EttOuKk

Goal

Master Java fundamentals so writing code becomes second nature. Build a strong foundation that makes every next topic easier.

Phase 2 – Problem Solving (2–4 Months)

Learn DSA

- Time & Space Complexity
- Arrays
- Strings
- Hashing
- Linked Lists
- Stacks & Queues
- Trees
- Binary Search Trees
- Heaps
- Graphs
- Dynamic Programming

 Solve **200–300 coding problems** on LeetCode or similar platforms.

Resource

 **DSA – Kunal Kushwaha**https://www.youtube.com/watch?v=rZ41y93P2Qo&list=PL9gnSGHSqcnr_DxHsP7AW9ftq0AtAyYqJ

Goal

Develop strong problem-solving skills and think like a software engineer. Learn to write efficient, optimized solutions with confidence.

Phase 3 – Development Tools (Parallel)

Learn

- Git
- GitHub
- IntelliJ IDEA / VS Code
- Maven
- Gradle

Goal

Work like a professional developer and manage projects efficiently. Understand the tools used in real software teams.

Phase 4 – Backend Development (4–6 Months)

Learn Spring Boot

- Spring Boot
- REST APIs
- Spring Data JPA
- Spring Security
- JWT Authentication
- Microservices (Basics)

Resource

 **Spring Boot – Programming with Mosh** <https://www.youtube.com/watch?v=gJrjgg1KVL4&t=975s>

Goal

Build production-ready backend applications using modern Java technologies. Understand how real-world web applications are developed.

Phase 5 – Database

Learn

- SQL
- MySQL / PostgreSQL
- Database Design
- Joins
- Indexing
- Transactions

Resource

 **SQL** https://youtu.be/JtaOmwnR6AM?si=x0Ql_bZWJ2kfMg3l

Goal

Design efficient databases and write optimized SQL queries. Store, retrieve, and manage application data with confidence.

Phase 6 – Build Projects

Build These Industry-Level Projects (Here's my top 10 projects)

- E-Commerce Backend
- Real-Time Chat Application
- Banking System API
- Task Management System
- Learning Management System (LMS)
- Job Portal
- Inventory Management System
- AI Resume Analyzer
- AI PDF Chat (RAG)
- Food Delivery Backend

Goal

Turn knowledge into real experience through practical projects. Build a portfolio that proves your skills to recruiters.

Phase 7 – AI Skills (Future Edge)

Learn

- Python Fundamentals
- Prompt Engineering
- OpenAI API
- LangChain
- RAG (Retrieval-Augmented Generation)
- AI Agents
- MCP (Model Context Protocol)
- AI Integration with Java

Goal

Stay ahead by combining software engineering with AI. Build intelligent applications that solve real-world problems.

Phase 8 – Career Preparation

Learn

- Resume
- GitHub Portfolio
- LinkedIn Profile
- Mock Interviews
- System Design (Basics)

- Docker Basics
- Deployment
- CI/CD (Basics)
- Apply for Internships & Software Developer Roles

Goal

Present your skills professionally and prepare for interviews. Start applying consistently until you land your first developer role.

Final Checklist

- ✓ Build **3–5 impressive projects**
- ✓ Keep your **GitHub active**
- ✓ Create a **professional portfolio**
- ✓ Practice **DSA regularly**
- ✓ Apply to jobs **consistently**

Final Message

Stay consistent, trust the process, and keep building every single day. Small progress compounds into extraordinary results.

Your first software developer job isn't a matter of luck—it's the result of disciplined learning, real projects, and relentless consistency. Keep building. Your breakthrough is closer than you think. 🚀